A shared-reset coupling outside every finite product construction

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Task 1. MathDB 339681 records Conjecture 19 of Nüsken and Pavliotis [1, pp. 337–338]. For generators of ergodic Feller semigroups, the conjecture asserts that one can choose decompositions

$$L_i = -\sum_{k=1}^{K_i} (A_k^i)^* A_k^i + B^i$$

so that every coupling operator is a sum

$$\Gamma = \sum_{\substack{i \neq j \\ 1 \le k \le K_i \\ 1 \le l \le K_j}} \alpha_{ijkl} A_k^i A_l^j, \tag{1}$$

with scalar coefficient functions on the product state space. We disprove the stated finite-sum construction. The obstruction holds for every choice of its finite families of scalar operators.

Theorem 1. *Let each of two marginal processes on the circle $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ reset, at rate one, to an independent uniform point of the circle. The coupling that resets both coordinates simultaneously to the same uniform point is a rich ergodic Feller coupling. Its coupling operator cannot have the form* (1) *for any finite K_1, K_2 and any decompositions of the marginal generators.*

Proof. Let π be normalized Lebesgue measure on $\mathbb{T}$, and define

$$(Pf)(x) = \int_{\mathbb{T}} f(z)\, d\pi(z), \qquad L = P - I.$$

Here P is a positive, constant-preserving projection of norm one on $C(\mathbb{T})$. Thus

$$S_t = e^{-t} I + (1 - e^{-t}) P \tag{2}$$

is a conservative, uniformly continuous Feller semigroup, with bounded generator L. It is precisely the stated reset process. Its invariant measure is uniquely π, and $S_t f$ converges uniformly to $\pi(f)$. Ergodic time averages follow directly from the strong law: successive reset positions are independent with law π, and the independent holding times are exponential with mean one, so the ratio of the accumulated rewards and holding times converges to $\pi(f)$. This applies to every bounded continuous observable and every initial point. Smooth functions are invariant under S_t and dense in $C(\mathbb{T})$. Since L is bounded, they are a core, and the process is rich.

The required marginal decomposition exists. Indeed, P is an orthogonal projection in $L^2(\pi)$, so with $A = I - P$ and $B = 0$,

$$-A^*A = -(I - P)^2 = P - I = L.$$

Our argument below excludes all other finite decompositions as well.

On $C(\mathbb{T}^2)$, put

$$(P_x F)(x,y) = \int_{\mathbb{T}} F(z,y)\, d\pi(z), \qquad (P_y F)(x,y) = \int_{\mathbb{T}} F(x,z)\, d\pi(z),$$

$$(QF)(x,y) = \int_{\mathbb{T}} F(z,z)\, d\pi(z).$$

The shared-reset generator and its coupling operator are

$$\overline{L} = Q - I, \qquad \Gamma = \overline{L} - (L_x + L_y) = Q - P_x - P_y + I. \tag{3}$$

The projection Q is positive and constant-preserving, so $\overline{S}_t = e^{-t}I + (1 - e^{-t})Q$ is again a conservative, uniformly continuous Feller semigroup. Its process resets the pair to (Z, Z), with $Z \sim \pi$, at each rate-one Poisson time. Each coordinate therefore has exactly the prescribed marginal process, including its path law. The unique invariant distribution is the diagonal measure $\Delta_{\#}\pi$, where $\Delta(z) = (z, z)$; the same renewal argument proves ergodicity.

All the operators in (3) are bounded in the supremum norm, and Γ is defined on all bounded Borel functions. It vanishes on functions of either coordinate alone. On $C(\mathbb{T}^2)$, $L_x + L_y + \Gamma = Q - I$ is the bounded Feller generator just constructed. Smooth functions on $\mathbb{T}^2$ are a core. Thus both conditions of the source's Definition 6 hold, including closability and the generator property. In fact $\Gamma \in \mathcal{G}^0$.

For any smooth f, g, (3) gives

$$\Gamma(f \otimes g)(x, y) = \int_{\mathbb{T}} f(z)g(z)\, d\pi(z) - g(y)\pi(f) - f(x)\pi(g) + f(x)g(y). \tag{4}$$

Now suppose (1) holds for finite K_1, K_2. Operators acting in different coordinates commute on simple tensors. Combining the terms with $(i, j) = (1, 2)$ and $(2, 1)$, we obtain

$$\Gamma(f \otimes g)(x, y) = \sum_{k=1}^{K_1} \sum_{l=1}^{K_2} \beta_{kl}(x, y)\, (A_k^1 f)(x)\, (A_l^2 g)(y). \tag{5}$$

Consequently, for any m test functions $f_1, \ldots, f_m$ and any fixed (x, y), the matrix

$$M(x, y) = \big[\Gamma(f_i \otimes f_j)(x, y)\big]_{i,j=1}^{m}$$

has a factorization $U(x)\beta(x, y)V(y)^{\mathsf{T}}$, and hence

$$\operatorname{rank} M(x, y) \le \min(K_1, K_2). \tag{6}$$

Choose $m > \max(K_1, K_2)$. Take a nonempty open arc O of the circle whose complement contains an open arc. In that complementary arc choose m disjoint smaller arcs, and smooth real bump functions f_i, one supported in each smaller arc, normalized by $\int_{\mathbb{T}} f_i^2\, d\pi = 1$. Then $f_i = 0$ on O and $\int_{\mathbb{T}} f_i f_j\, d\pi = \delta_{ij}$. Equation (4) therefore yields

$$M(x, y) = I_m \qquad \text{for every } (x, y) \in O \times O,$$

contradicting (6). The contradiction also works if the factors are interpreted as L^2 representatives and the identities hold almost everywhere: only finitely many factors and test functions occur, so choose a point of the positive-measure set $O \times O$ outside their exceptional null sets. This completes the disproof. □

Scope and verification. The final source permits locally compact Polish state spaces with a smooth structure; the compact circle satisfies this condition. Conjecture 19 is stated for ergodic Feller processes generally. The following Proposition 20 treats the narrower continuous-path diffusion case, which this jump-process example does not contradict. The conclusion concerns the finite index ranges displayed in equations (34), (37), and (39); an infinite expansion or an auxiliary noise-space integral would be a different construction. The accompanying exact checks verify projection, generator, marginal, and rank identities in finite reset models. The arbitrary-rank argument on smooth functions above supplies the full proof.

Task 2. Formulate and prove an exhaustion theorem that permits countably many operators or integration over an auxiliary noise space, with explicit convergence and domain conditions.

Task 3. Characterize jump kernels whose coupling operators admit a finite product representation of the form (1).

Task 4. Determine the minimum numbers of marginal factors needed to represent all coupling operators for finite-state irreducible Markov chains.

Task 5. Develop finite-rank approximations of shared-reset couplings and quantify their effect on invariant measures and asymptotic variances.

References

[1] N. Nüsken and G. A. Pavliotis, *Constructing Sampling Schemes via Coupling: Markov Semigroups and Optimal Transport*, SIAM/ASA J. Uncertainty Quantification **7**(1) (2019), 324–382, doi:10.1137/18M119896X. Conjecture 19, equations (34)–(39), pp. 337–338; Definition 6, p. 332; general setting, pp. 330–331.